

ETHAN YOUNGIN SHIN

Email: youngin@mit.edu

Address: 15 Vassar Street (Building 48), Cambridge MA 02139

RESEARCH INTERESTS

Atmospheric Boundary Layer, Turbulence, Uncertainty Quantification, Bayesian Inference, Environment-Energy System Interactions, Engineering for Global Development

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA, United States

Sep. 2022 - Present

Ph.D. Candidate in Civil and Environmental Engineering

- Relevant coursework: Fluid Mechanics, Fundamentals of Turbulence, Atmospheric Flow and Wind Energy, Numerical Methods for Stochastic Modeling and Inference, Introduction to Numerical Methods, Air-Sea Interaction

Yonsei University, Seoul, Republic of Korea

Mar. 2016 - Aug. 2022

B.S. in Mechanical Engineering

Honors: *magna cum laude*

RESEARCH EXPERIENCE

Howland Lab (Prof. Michael Howland), Massachusetts Institute of Technology

Research Assistant

Sep. 2022 - Present

- Improving atmospheric boundary layer modeling using Bayesian model calibration and uncertainty quantification
- Developing an optimal experimental design-based sensor placement framework for wind resource characterization

Innovative Design and Integrated Manufacturing Lab (Prof. Sung-Hoon Ahn), Seoul National University

Research Intern

Dec. 2020 - Feb. 2022

- Conducted field demonstration of a polling-based energy management system with an off-grid solar power plant in Tanzania
- Investigated feasibility of a smart electric mobility operating system for integration with off-grid solar power plants in Tanzania

Multiphysics Energy System Laboratory (Prof. Jongsup Hong), Yonsei University

Research Intern

Jul. 2020 - Aug. 2020

- Conducted a thermofluidic analysis of a solid oxide fuel cell (SOFC) cooling system using ANSYS Fluent
- Designed and tested the effects of rib turbulators on SOFC interconnect fuel channels

PUBLICATIONS

E. Shin, M. Howland*. “Accelerated Bayesian calibration and uncertainty quantification of RANS turbulence model parameters for stratified atmospheric boundary layer flows.” (*In review*).

E. Shin, X. Yang, M. Howland*. “Addressing grid convergence and log-layer mismatch in wall modeled large eddy simulations of geophysical flows over rough surfaces and canopies.” (*In review*).

E. Shin, M. Chan, J. Wang, T. Zahtila, C. Gorle, G. Iaccarino, M. Howland*. “Multifidelity modeling and uncertainty quantification of heterogeneous roughness.” *Center for Turbulence Research, Proceedings of the Summer Program 2024*. (2024)

H. Rhee, H. Im, F. Manongi, **Y. Shin**, H. Song, W. Jung, S. Ahn*. “Smart Electric Mobility Operating System Integrated with Off-Grid Solar Power Plants in Tanzania: Vision and Trial Run.” *Journal of Appropriate Technology*, Vol. 7, No. 2, 127-135. (2021)

CONFERENCE PRESENTATIONS

E. Shin, M. Howland*. “Bayesian Inversion to Quantify Parameter Uncertainty in RANS Turbulence Models for Stratified Atmospheric Boundary Layer Flows.” *AMS 25th Symposium on Boundary Layers and Turbulence*. (2025)

E. Shin, M. Howland*. “Accelerated uncertainty quantification of RANS turbulence models for stratified atmospheric boundary layer flow.” *American Geophysical Union*. (2024)

E. Shin, M. Chan, J. Wang, T. Zahtila, C. Gorle, G. Iaccarino, M. Howland*. “Multi-fidelity modeling and uncertainty quantification of heterogeneous roughness.” *Bulletin of the American Physical Society*. (2024) (*Invited*)

E. Shin, M. Howland*. “Uncertainty quantification of RANS turbulence model parameters with ensemble Kalman methods and machine learning.” *Bulletin of the American Physical Society*. (2023)
H. Im, **Y. Shin**, G. Jung, S. Ahn*. “Field demonstration of polling-based energy management system in rural Tanzania.” *International Conference on Appropriate Technology*. (2021)

AWARDS AND SCHOLARSHIPS

CEE Mathworks Fellowship, Massachusetts Institute of Technology (Fall 2023, Spring 2024)
Louis Berger Fellowship, Massachusetts Institute of Technology (Spring 2023)
Schoettler Fellowship, Massachusetts Institute of Technology (Fall 2022)
Highest Honors, Yonsei University (Spring 2021)
Honors, Yonsei University (Fall 2016, Spring 2019, Spring 2020, Fall 2020)
Merit-based Scholarship, Yonsei University (2019, 2020, 2021, 2022)
National Scholarship, Korea Student Aid Foundation (2016, 2019, 2020)
Lee Choong Kon Pay Forward Scholarship (Spring 2019)

SKILLS

Languages	English, Korean, Russian
Programming	Julia, MATLAB, Python, Fortran, C, C++
Specialized	OpenFOAM, Large Eddy Simulation, Weather Research & Forecasting (WRF) Model

TEACHING EXPERIENCE

Guest Lecturer
Leaders in Industry-University Cooperation 3.0 *Jul. 2023 - Present*
• Lectured on how to model wind in the atmospheric boundary layer to undergraduate students from Korean universities (Jeonbuk National University, Sookmyung Women’s University)

Participant
MIT Teaching & Learning Lab *Feb. 2023 - Dec. 2023*
• Completed all four tracks of the Grad Teaching Development Tracks (equivalent to Kaufman Teaching Certificate Program), a set of workshops intended for early- and mid-program graduate students interested in improving their teaching skills

Part-time Teacher
ASAP Academy *Jul. 2019 - Aug. 2020*
• Taught high school math, physics, and English grammar to classes of up to 7 students

VOLUNTEER EXPERIENCE

Volunteer <i>Loaves and Fishes Meal Program</i>	<i>Dec. 2022 - Sep. 2024</i>
Volunteer <i>Yonsei Habitat for Humanity</i>	<i>Mar. 2020 - Jun. 2022</i>
Student Volunteer <i>Incheon Support Center for Foreign Workers</i>	<i>Mar. 2016 - Jun. 2016</i>

LEADERSHIP EXPERIENCE

Panel Director <i>MIT Energy Conference</i>	<i>Oct. 2023 - Mar. 2024</i>
President <i>Charles River Running Crew, MIT KGSA</i>	<i>June. 2023 - June. 2024</i>
15th Class President <i>Yonsei Habitat for Humanity</i>	<i>Mar. 2020 - Jun. 2021</i>
Student Ambassador <i>Yonsei-Waseda Exchange Program</i>	<i>Jul. 2016, Feb. 2019</i>
Police Bus Driver <i>Auxiliary Police, Republic of Korea (Military Service)</i>	<i>Apr. 2017 - Dec. 2018</i>